IRRIGATION WATER MANAGEMENT · NOTE SP03

Infiltration and Evapotranspiration — Infiltration, Evaporation, Transpiration and Consumptive Use

Course topics: IDE5.5 L13, SWCE3.5 L9, IDE-501 T1, IDE-512 T7

Infiltration

- Infiltration is the downward entry of water into the soil through its surface; the rate at any instant (cm per hour) is the **infiltration rate**. It is high at first, falls as the soil wets and finally becomes nearly constant — the **basic infiltration rate**.
- Knowing field infiltration rates is essential for designing systems and for working out how long to irrigate to a desired depth.
- **It varies with** texture, structure, initial moisture and the salt content of the water. The least permeable layer in the profile controls it.
- **Surface sealing:** raindrop impact and flowing water form a thin compact crust as fine particles block the pores.
- **Typical rates:** over 25 cm/hr in very sandy soils; approaching zero in clays — reduced further by alkalinity, salinity and poor structure.

Measuring infiltration

- **Direct:** measure the water applied and the water running off; the difference is the infiltration (cm of water per hour).
- **Concentric-cylinder infiltrometers** where direct measurement is impractical: set several in the area, add water gently without disturbing the soil, keep a constant head and read at frequent intervals with a point or hook gauge. The outer ring (**buffer pond**) limits lateral spread — keep the water levels in both rings about equal.
- **Report** as depth infiltrated in a given time, rate during a definite period, or time to infiltrate a given depth; plot accumulated infiltration and average rate against time.

Evaporation

Evaporation transfers liquid water to the atmosphere as vapour; in agriculture it means the total loss from an area over a time, as mass or volume per unit area or as a depth of water. **Transpiration** is the loss from plant surfaces only; **evapotranspiration** is the sum of the two.

- It rises with water and air temperature, dry air and wind; coastal areas evaporate less, hot dry windy regions more.
- Estimates refer to a free water surface; most formulae rest on Dalton's theory, and any formula must be checked against field observations.

Evaporation pan

- The common pan is 120 cm in diameter, of mild steel painted white, holding 17–20 cm of water, set on a wooden grate about 15 cm above the ground and covered with a wire screen.
- Read the loss with a hook gauge, regularly over months.

- A pan evaporates more than a large free-water surface, so multiply by a pan coefficient of about 0.7.

> **Pan depth.** The notes give the pan as 20 cm deep while holding 17–20 cm of water. The standard USWB Class A pan (adopted in India) is about 122 cm across and **25 cm deep**, with water kept 5–7.5 cm below the rim — which is where the 17–20 cm water depth comes from.

Evaporation from land depends on how saturated the soil is, air and soil temperature, humidity, wind and vegetative cover: a very wet soil evaporates like free water; as it dries, evaporation falls and is confined to shallow layers.

Transpiration

- The discharge of water vapour from the leaves and stems of living plants; it governs how much and how often to irrigate. It ranges from almost zero (young seedlings, cold cloudy weather) to its maximum (hot windy weather, luxuriant growth); rapidly growing leafy crops may transpire about 5 cm of water per week at peak growth.
- **Depends on** available soil moisture, temperature, humidity, wind, intensity and duration of sunlight, growth stage, species and leaf characteristics.
- **Transpiration ratio** — water transpired per unit of dry matter produced; formerly used as a measure of water-use efficiency.

Consumptive use (evapotranspiration)

- The sum of transpiration and evaporation — water used by plants plus water evaporated from soil, water surfaces and foliage — expressed in hectare-centimetres per hectare or cm depth. It is the water an ideal irrigation system must supply to a crop, field or farm.
- **Through the season** it is low at first, rises with foliage and temperature, peaks at fruiting and declines towards maturity; the total is the **seasonal consumptive use**.
- **Peak-period use rate** — the average daily use over the 6–10 days of highest consumptive use, when vegetation is abundant, the crop is producing harvest and temperatures are high. It is the design rate for irrigation planning.

Measuring consumptive use	How
Lysimeters	Grow the crop in large tanks and weigh them periodically; weight change = consumptive use. Limited by artificial conditions, tank size and soil
Field plots	More dependable: irrigate without runoff or deep percolation, plot yield against water used and find the optimum
Soil-moisture studies	Measure moisture before and after irrigation, compute daily extraction and get seasonal use from the curve; needs uniform soil and a deep water table
Evaporation pan	ET is closely related to pan evaporation — useful for scheduling

$$\text{Evapotranspiration} = \text{pan evaporation} \times \text{crop coefficient}$$

- **Crop coefficient (crop factor):** relates crop ET to the reference (here pan) evaporation; it varies with crop stage, humidity, wind, crop and location — typically 0.9 – 1.2 for common crops.
- **Actual evapotranspiration** — the plants' actual water uptake, limited by the available soil moisture.

- **Potential evaporation** — evaporation when all surfaces are wet with no biological or soil-moisture limit; depends on the weather, surface albedo and vegetation type and density.

Definition fixed. The notes define the crop coefficient as "maximum ET ÷ actual ET". The standard definition (FAO-56) is Kc = crop evapotranspiration ÷ reference evapotranspiration (ETc = Kc × ET_0); the pan form above is the same idea with pan evaporation as the reference.

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) lessons on soil–water–plant relations, evapotranspiration, irrigation efficiencies, and surface, sprinkler and drip irrigation. FAO Irrigation & Drainage Paper 56 (crop coefficient). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The nearly constant infiltration rate reached after long wetting is the

(a) initial rate
(b) basic infiltration rate
(c) percolation rate
(d) hydraulic conductivity

Q2. The outer ring of a double-ring infiltrometer serves as a

(a) measuring ring
(b) buffer pond
(c) rain gauge
(d) drain

Q3. Pan coefficient used to convert pan evaporation is about

(a) 0.3
(b) 0.5
(c) 0.7
(d) 1.2

Q4. Peak-period use rate is the average daily use over the ___ of highest use

(a) 1–2 days
(b) 6–10 days
(c) 30 days
(d) whole season

Q5. Consumptive use is the sum of

(a) infiltration and runoff
(b) evaporation and transpiration
(c) percolation and seepage
(d) rainfall and irrigation

Q6. Very sandy soils can infiltrate more than

(a) 0.5 cm/h
(b) 2.5 cm/h
(c) 25 cm/h
(d) 250 cm/h

Answers

Q1 — (b) basic infiltration rate
Q2 — (b) buffer pond
Q3 — (c) 0.7
Q4 — (b) 6–10 days
Q5 — (b) evaporation and transpiration
Q6 — (c) 25 cm/h